

Generation Interconnection System Impact Study Report

For Queue Project Q19-1217

Fumarole 138 kV

April 29, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1217. The Interconnection Customer proposes a 39.3 MW battery energy storage facility interconnecting at the Fumarole 138 kV substation in the Virginia area. The request was received on June 02, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 326 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 26 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,253,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Clark County 345/138 kV TX	Artesia Country Club to Clark County 345kV	121.8	512	9.5
Columbus City Substation 138/1 kV Transformer	Kearneyville to Clark County 345kV	116.4	495	9.3
Air Force Base 345-138 kV Auto	Air Force Base to Tiger Bay 345kV	113.8	541	5.9
Clark County to Shelton 138kV L	Artesia Country Club Substation to Clark County 345kV Line	113.2	246	6.1
Le Grand to West Winchester 138kV	International Paper Valliant Ok to West Winchester 138kV	112.9	257	8.8
Opequon to Lcec No. 5 138kV Line	Clark County to Kearneyville 345kV	110.2	348	8.3
N Mesa to Opequon 138kV	Clark County to Kearneyville 345kV	109.9	359	8.3
West Winchester to Santa Rosa Energy Ctr 138kV Ckt 1	Santa Rosa Energy Center to W Winchester 138kV	109.1	303	6.1
W Winchester to Santa Rosa Energy Ctr Substation 138kV Cir. 2	W Winchester to Santa Rosa Energy Ctr 138kV	109.1	303	6.1
Abbotsford to Le Grand Substation 138kV	W Winchester to International Paper Valliant Ok 138kV Line	107.9	288	8.8
Fie Channels to Shelton 138kV L	Artesia Country Club Substation to Clark County 345kV	106.5	239	6.1
Terror Lake Microgrid to Kitimat 138kV	Clark County to Kearneyville 345kV	105.8	283	5.7
Kitimat to Cascade Steel 138kV	Clark County to Kearneyville 345kV	105.6	296	5.7
Loughridge to Firewheel 138kV	American to Urban 345kV	105.6	245	5.5
Cow to Feagan'S Mill 138kV	Clark County to Kearneyville 345kV	105.4	315	5.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Yahill to Cow 138kV	Clark County to Kearneyville 345kV	105.3	328	5.7
3 Mile Knoll Substation to Yamhill 138kV	Kearneyville to Clark County 345kV	105.3	335	5.7
Edens Ridge to Firewheel 138kV	American to Urban 345kV	105.2	349	7.7
Feagan'S Mill Substation to Cascade Steel 138kV L	Clark County to Kearneyville 345kV	105.2	310	5.7
Columbus City 345/1 kV Autotransformer	Clark County to Kearneyville 345kV	104.7	563	9.3
Morbihan 345/138 kV TX	Ridgeley to Spindle 345kV	103.8	611	7.2
Loretto to Black Oak 138kV	Ridgeley to Spindle 345kV	103.0	481	5.6
North Mesa to Columbus City 138kV	Clark County to Kearneyville 345kV	102.9	323	6.7
Edens Ridge to Midset 138kV	Urban to American 345kV L	101.4	333	7.7
Loretto to Crockett 138kV	American to Urban 345kV line	101.2	578	8.3
Chambersburg 500-345 kV TX	E Devon to Thiele Kaolin Reedy Crk 345kV	101.0	1,030	6.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Clark County 345/138 kV TX	Replace transformer with higher-rated unit	\$20,928,000	4% (\$735,000)
Columbus City 138 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$4,940,000	2% (\$115,000)
Air Force Base 345-138 kV Auto	Replace transformer with higher-rated unit	\$14,200,000	2% (\$289,000)
Clark County to Shelton 138kV L	Rebuild line with higher-capacity conductor	\$952,000	2% (\$20,000)
Le Grand to West Winchester 138kV	Rebuild line with higher-capacity conductor	\$4,139,000	34% (\$1,419,000)
Opequon to Lcec No. 5 138kV Line	Rebuild line with higher-capacity conductor	\$5,938,000	4% (\$223,000)
N Mesa to Opequon 138kV	Reconductor line	\$719,000	4% (\$27,000)
West Winchester to Santa Rosa Energy Ctr 138kV Ckt 1	Reconductor line	\$2,153,000	24% (\$516,000)
W Winchester to Santa Rosa Energy Ctr Substation 138kV Cir. 2	Reconductor line	\$2,291,000	24% (\$549,000)
Abbotsford to Le Grand 138kV	Reconductor line	\$3,412,000	34% (\$1,170,000)
Five Channels to Shelon 138kV L	Reconductor line	\$762,000	2% (\$16,000)
Kitimat to Terror Lake Microgrid 138kV	Reconductor line	\$1,384,000	10% (\$137,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cascade Steel to Kitimat 138kV line	Reconductor line	\$613,000	10% (\$61,000)
Firewheel to Loughridge 138kV	Reconductor line	\$5,811,000	100% (\$5,811,000)
Cow to Feagan'S Mill 138kV	Reconductor line	\$1,229,000	10% (\$122,000)
Yahill to Cow 138kV	Reconductor line	\$863,000	10% (\$86,000)
Yamhill to 3 Mile Knoll 138kV	Reconductor line	\$1,424,000	10% (\$141,000)
Edens Ridge to Firewheel 138kV	Reconductor line	\$1,923,000	100% (\$1,923,000)
Feagan'S Mill Substation to Cascade Steel 138kV L	Reconductor line	\$1,299,000	10% (\$129,000)
Columbus City 345 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$24,106,000	2% (\$562,000)
Morbihan 345/138 kV TX	Replace transformer with higher-rated unit	\$14,037,000	9% (\$1,204,000)
Loretto Substation to Black Oak 138kV	Reconductor line	\$3,186,000	9% (\$274,000)
North Mesa to Columbus City 138kV	Reconductor line	\$455,000	2% (\$11,000)
Edens Ridge to Midset 138kV	Reconductor line	\$2,585,000	100% (\$2,585,000)
Crockett to Loretto 138kV L	Reconductor line	\$1,042,000	6% (\$65,000)
Chambersburg 500/345kV TX	Replace transformer with higher-rated unit	\$24,013,000	0% (\$63,000)
	Total allocated to project		\$18,253,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.